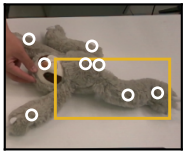
A PhysTwin real transfer: lift → push

Selected successful pair · full-interaction error: 58.23 → 42.78 mm · camera 0, final frame

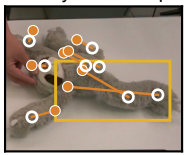
Recorded + manual tracks





Same highlighted landmark

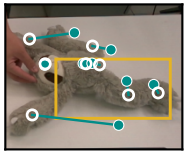
Frozen PhysTwin adaptation





3D error: 206.1 mm

Ours: frozen neural adapter





3D error: 31.6 mm

B SoftVTBench: improved placement after slip

Selected best episode-level gain · 22.63 → 17.40 mm · unchanged shape, better position

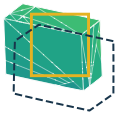
Simulator reference





Same material point

Mechanics + contact





Point error: 18.5 mm

Ours: slip-aware prediction





Point error: 13.6 mm

Point error (mm), common scale →

